

2. Tomos and Jerry wish to determine the material from which a ball bearing is made. They both decide to determine the density of the metal of the ball bearing. However, they choose different methods.

Tomos' method:

Tomos uses digital callipers to measure the diameter of the ball bearing. He then uses a digital balance to determine the mass of the ball bearing. He obtains the following values:

Diameter of ball bearing = 18.76 ± 0.02 mm

Mass of ball bearing = 26.3 ± 0.5 g

Jerry's method:

Jerry measures the volume directly by submerging the ball bearing in water in a measuring cylinder. He also uses the same digital balance to determine the mass of the ball bearing and obtains the following values:

Volume of ball bearing = 3.4 ± 0.1 cm³

Mass of ball bearing = 26.3 ± 0.5 g

(a) **Using Tomos' values:**

- (i) Calculate the volume of the ball bearing in cm³ and show that its **percentage** uncertainty is approximately 0.3%. [3]

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- (ii) Hence calculate the density of the ball bearing along with its **absolute** uncertainty, giving your answer to an appropriate number of significant figures. [4]

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- (b) Jerry believes that, since he is measuring the volume directly, the absolute uncertainty in his value of density will be less than that found by Tomos. Determine whether or not he is correct. [2]

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- (c) The table gives the density of some common metals and alloys.

Metal	Density/g cm ⁻³
tin	7.3
stainless steel	7.5
iron	7.9
brass	8.3

Tomos and Jerry use this information to determine the metal from which the ball bearing is made. Explain how Tomos' conclusion differs from Jerry's. [2]

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